A ring of charge as a Kron machine

Do Kron's machine equations model the ring electron? — register entry H21

kronspice simulation report · 2026-11-07

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Summary

A closed ring of circulating charge is written as one of Gabriel Kron's machines: a winding whose current is i = q·dθ/dt/2π, with the ring's mass entering Kron's metric tensor as an inductance. The example first checks this machine against Kelvin's 1899 charged ring, solved independently with the Lorentz force, then fixes a ring by the electron's charge and spin and compares its moment and g-factor with the measured electron. The hypothesis (H21) is that the electron *is* such a ring; the standard prediction beside every row is the Dirac electron with QED (g = 2.002 319 304 36, μ = 1.001 16 μ_B).

- **Twin test passes.** Kelvin's ring as a three-winding machine gives moments of momentum −/+1.11207×10^{-10} kg m²/s (= eΦ/2π), the fluid at Larmor's rate 4.44827×10^{-6} rad/s and kinetic energy 4.94677×10^{-16} J, against 4.94683×10^{-16} J from the Lorentz-force bench.
- **Torque from the metric is −μB sin θ.** The tilting ring's period is 19.873 s against the pendulum value 4K(k)/ω_0 = 19.872 s (2×10^{-5}); ledger mismatch 6×10^{-10} J.
- **A ring whose mass circulates with its charge has g ≤ 1:** g = 1/(1 + L·e²/4π²·m·a²), so it precesses at Larmor's eB/2m, half the electron's spin precession.
- **Only a massless ring reaches the measured moment** at S = ħ/2, at radius 8.86×10^{-15} m, with v = 43.6 c and a magnetic energy of 475 $m_e c^2$.
- **Every ring is far larger than the 10^{-18} m scattering bound** on the electron's size.
- **Status: bounded.** The machine reproduces the classical ring exactly and inherits its obstacles (g ≤ 1 or v > c).

What is being tested

Hypothesis H21 (user hypothesis, 2026-10-31): a closed ring of circulating charge is a one-winding Kron machine, with self-inductance L, the circulating mass as rotor inertia and an external field entering as speed voltages; voltage and current are the pressure and flow of energy at that scale, and the electron is such a ring. The established part is Kron's own: his eq. 11 (*Equivalent Circuits of Electric Machinery*, 1951, Epilogue p. 259) states that with e = R = 0 the machine equation is the motion of a charged particle in an electromagnetic field.

Standard prediction: the Dirac electron with QED, g = 2.002 319 304 36 and μ = 1.001 16 μ_B; for a classical ring, g = 1 when mass circulates with charge, a Compton-scale radius for μ_B at v ≤ c, and no stable state. Scattering bounds the electron below 10^{-18} m.

The model

For a ring of charge q and mass m turning at dθ/dt, the current is i = q·dθ/dt/2π, so the electrical and the mechanical coordinate are the same axis. The kinetic energy ½·m·a²·dθ/dt² = ½·(m·a²·(2π/q)²)·i² is then a magnetic energy on that axis, and the inertia enters Kron's metric as an inductance m·a²·(2π/q)², beside the ring's own self-inductance L. Kron defined the metric as "all the self- and mutual inductances and the rotor inertia" (1951, Epilogue p. 254, eqs. 3–4). Everything uses the N machine element (device/machine.go) unchanged.

quantity	formula / value	source
ring self-inductance	L = μ_0·a·(ln(8a/r) − 7/4)	Maxwell, *Treatise* Art. 706
ring capacitance	thin-ring formula	Smythe §5.17
inertial inductance	m·a²·(2π/q)²	Kron's metric
coil–ring mutual	em.MutualCircles (elliptic form)	Maxwell

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quantity	formula / value	source
twin test: coil	6.201×10^{-7} H	printed metric
twin test: coil–ring mutual	3.494×10^{-8} H	printed metric
twin test: tube, fluid inertial inductance	9.87×10^{8} H each	printed metric
twin test: ring's own L	3.1×10^{-7} H (dropped as in Kelvin, or kept: no change)	

1. Twin test: Kelvin's ring in Kron's frame (standard)

Kelvin's ring (tube +e, fluid −e, each 10 g, radius 5 cm; coil of 20 kA-turns, radius 10 cm, switched on over 1 s, 5 cm below) becomes a three-winding machine: coil, tube and fluid. Tube and fluid windings are closed through 0 V sources (frictionless), so their flux linkages are conserved.

quantity	Kron machine	Lorentz-force bench
flux through the ring	6.98732×10^{-4} Wb (Maxwell elliptic)	6.98736×10^{-4} Wb (Biot–Savart)
moment of momentum, tube	-1.11207×10^{-10} kg m²/s	-1.11207×10^{-10}
moment of momentum, fluid	$+1.11207\times10^{-10}$	$+1.11207\times10^{-10}$
Kelvin's eΦ/2π	1.11207×10^{-10}	
fluid angular velocity	4.44827×10^{-6} rad/s	4.44829×10^{-6}
Larmor $e\cdot B_{avg}/2w$	4.44827×10^{-6}	
kinetic energy	4.94677×10^{-16} J	4.94683×10^{-16} J
source work against the ring	= kinetic + mutual-field energy (-9.89×10^{-16} J), residual 1.5×10^{-13}	residual 8×10^{-16}

The 1.1×10^{-5} difference in kinetic energy is the bench's Biot–Savart flux against Maxwell's elliptic value used here. The engine's own ledger also closes (source 124.028 J, machine's ½iᵀLi 124.02 J, mismatch 7.8×10^{-3} J over the ramp's breakpoints), but it is dominated by the coil's own 124 J of field energy, which is why the bench's bookkeeping is reported.

2. Torque from the metric (standard)

The same ring, spinning at 1000 rad/s (current 0.159 mA, moment 1.25×10^{-6} A·m²), tilts about a diameter in 1 T. In Kron's frame I, M(θ) = B·π·a²·cos θ couples a field winding and the ring, and the tilt inertia ½·m·a² is a capacitor on the mechanical port. Kron's torque $\frac{1}{2}\cdot i^{T}\cdot(\partial L/\partial\theta)\cdot i$ is −μ·B·sin θ. Released at 0.05 rad, the oscillation has period 19.873 s against the pendulum value $4K(k)/\omega_0$ = 19.872 s (2×10^{-5}). The ring current drifts by 6×10^{-11} (flux conservation) and the ledger mismatch is 6×10^{-10} J.

The ring current has no harmonics (a uniform ring's field is static). The tilt is a nonlinear pendulum, and the solver's ω(t) carries exactly the pendulum's odd series (Jacobi nome $q = e^{-\pi K'/K}$, $k = \sin\theta_0/2$), measured by leak-free Fourier sums over two periods:

θ_0	fundamental	ω_3/ω_1 run	exact	ω_5/ω_1 run	exact	ω_2/ω_1
0.05 rad	0.05032 Hz	3.906×10^{-5}	3.907×10^{-5}	2×10^{-8}	1.5×10^{-9}	2×10^{-8}
1 rad	0.0472 Hz	0.01658	0.01658	2.70×10^{-4}	2.71×10^{-4}	2×10^{-8}
2.5 rad	0.03063 Hz	0.1597	0.1598	0.0225	0.0225	8×10^{-7}

Even harmonics are absent by the symmetry θ → −θ.

3. The electron as the ring (hypothesis)

Charge e; the total angular momentum of the metric is set to ħ/2 and the rest follows (core radius a/100). KE is the kinetic energy, ½LI² the magnetic energy and W_{es} the electrostatic energy, each in units of $m_e c^2$.

ring	a [m]	v/c	μ/μ_B	g	KE	½LI²	W_{es}
Compton ring, electron mass circulating	3.86×10^{-13}	0.494	0.494	0.989	0.122	0.0014	0.0078
Half-Compton ring, mass circulating	1.93×10^{-13}	0.978	0.489	0.978	0.478	0.011	0.0155

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ring	a [m]	v/c	μ/μ_B	g	KE	½LI²	W_{es}
Compton ring, massless (field inertia)	3.86×10^{-13}	43.6	43.6	87.2	0	10.9	0.0078
Classical-radius ring, massless	2.82×10^{-15}	43.6	0.318	0.637	0	1490	1.06
Massless ring fitted to the measured moment	8.86×10^{-15}	43.6	1.001	2.002	0	475	0.338
Measured (Dirac + QED)	**$< 10^{-18}$ (scattering)**		**1.00116**	**2.00232**			

- A ring whose mass circulates with its charge has $g = 1/(1 + L\cdot e^2/4\pi^2\cdot m\cdot a^2) \le 1$: its precession in a field is Larmor's eB/2m, half the electron's spin precession.
- Only a massless ring (inertia from its own field) reaches the measured moment at S = ħ/2, at 8.9 fm with v = 43.6 c and a magnetic energy of 475 $m_e c^2$. Every massless thin ring with S = ħ/2 moves at $v/c = \pi/(2\alpha(\ln(8a/r) - 7/4))$, fixed by the fine-structure constant α; at v ≤ c ($a \ge \hbar/2m_e c$) its moment is far below μ_B.
- Every ring radius lies far above the 10^{-18} m scattering bound.

Natural frequencies of the rings (S = ħ/2, B = 1 T)

ring	rotation of the charge	precession (ESR)	tilt libration	LC self-resonance	wave once round at c
Compton ring, mass circulating	6.1×10^{19} Hz	13.8 GHz	1.3×10^{15} Hz	2.3×10^{19} Hz	1.24×10^{20} Hz
Half-Compton ring, mass circulating	2.4×10^{20}	13.7 GHz	2.6×10^{15}	4.6×10^{19}	2.47×10^{20}
Compton ring, massless	5.4×10^{21}	1.22 THz	—	2.3×10^{19}	1.24×10^{20}
Classical-radius ring, massless	7.4×10^{23}	8.9 GHz	—	3.1×10^{21}	1.7×10^{22}
Massless ring fitted to the moment	2.4×10^{23}	28.0 GHz	—	1.0×10^{21}	5.4×10^{21}
Measured / reference		**28.025 GHz (ESR)**			**1.236×10^{20} Hz ($m_e c^2/h$)**

- **Precession** is the spin's magnetic-resonance frequency g·eB/4πm, measured in examples/kron_nmr; only the fitted massless ring reaches the measured value, by construction.
- **Rotation** is the charge's turning rate, the current's fundamental: about half the Compton frequency for the Compton ring (v ≈ c/2).
- **Tilt libration** $\sqrt{(\mu B/J)}/2\pi$ is optical (10^{15} Hz) for an electron-mass ring in 1 T.
- **Wave once round at c** equals the Compton frequency $m_e c^2/h$ for the Compton ring and $2m_e c^2/h$ (Zitterbewegung) for the half-Compton ring: identities of the chosen radius, not predictions.
- There is no radial breathing mode: the model has no restoring force for the ring's radius.
- Kelvin's 5 cm ring of part 2 (1 μC, 10 g, 1000 rad/s, 1 T): rotation 159 Hz, precession $eB/4\pi m = 8\times10^{-6}$ Hz, libration 0.0503 Hz (the 19.9 s period), LC self-resonance 177 MHz.

Against existing measurements

quantity in the model	measured value	match?
Precession of a ring with circulating mass, eB/4πm = 14.0 GHz/T (g = 1)	Normal Zeeman effect: orbital moments precess at Larmor's eB/4πm, $g_L = 1$ (Zeeman 1897; Larmor 1897)	**Yes** for orbital moments; not the electron's spin (g = 2.0023).
Precession of the fitted massless ring, 28.02 GHz/T	Electron spin resonance; g/2 = 1.001 159 652 180 59(13) (Fan et al. 2023; Hanneke et al. 2008)	By construction only; needs v = 43.6 c.
Pendulum period $4K(k)/\omega_0$ and odd harmonics of the tilt	Large-amplitude pendulum measurements agree with the elliptic-integral period to about 10^{-4} (Nelson & Olsson 1986)	**Yes**, for the classical pendulum.
Wave once round the Compton ring, 1.236×10^{20} Hz; $2m_e c^2/h$ for the half-Compton ring	Identity of the radius. A channeling dip at $m_e c^2/h$ was reported (Gouanère et al. 2008), not reproduced; Zitterbewegung simulated with a trapped ion (Gerritsma et al. 2010), not seen in an electron	No measurement on the electron.
Charge rotation, tilt libration, LC self-resonance, modal series n·c/2πa	None: a uniform ring current radiates nothing at these frequencies	No measurement.

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The two genuine matches are classical: the ring's precession is the orbital Larmor frequency of the normal Zeeman effect, and its tilt is the measured large-amplitude pendulum. Neither bears on the electron's spin, which is what the g ≤ 1 bound says.

4. Figures

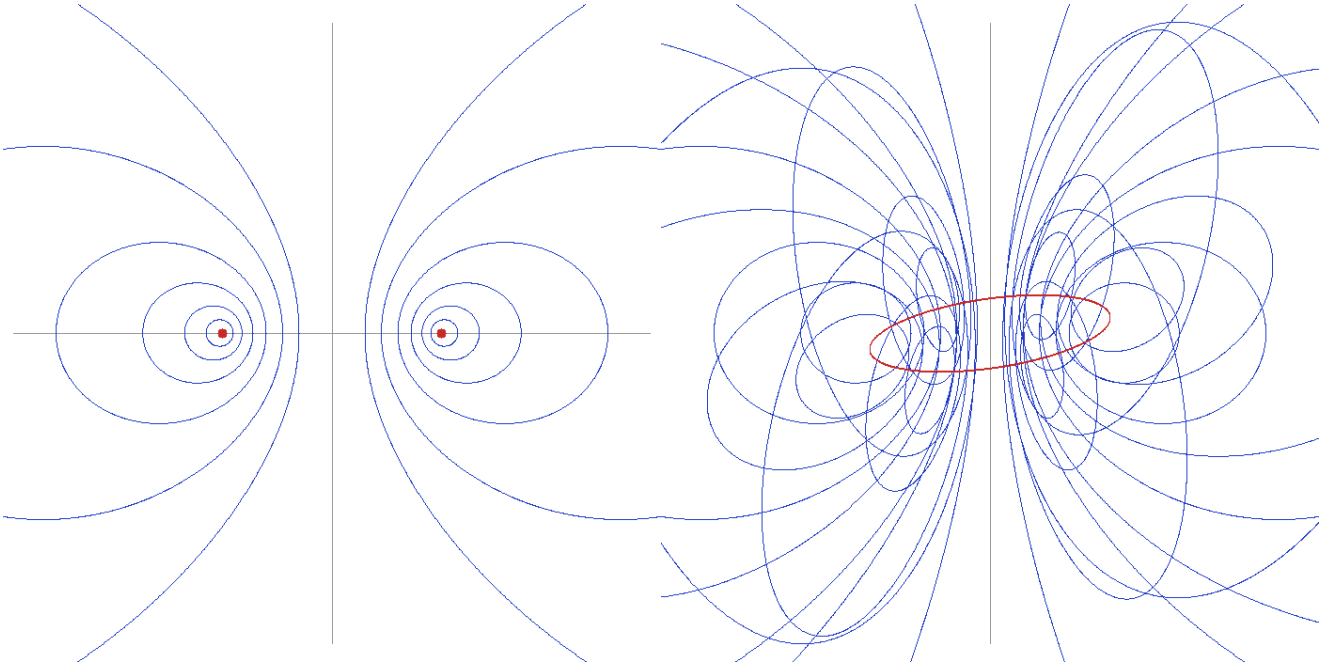


Figure 1. Field lines of the ring current from Biot–Savart. Left: the meridional plane, the wire as two red dots and the axis vertical. Right: oblique view at eight azimuths, the ring in red. Every line lies in a meridional plane and threads the ring once: dipole far field, near field circling the wire. Lines started at 0.3, 0.6 and 0.88 of the radius close to 10^{-9} (TestFieldLinesClose).

A single ring current cannot produce field lines that wind around a sphere, as drawn in some toroidal electron pictures; that needs a current component along the ring's axis, a second winding in Kron's frame.

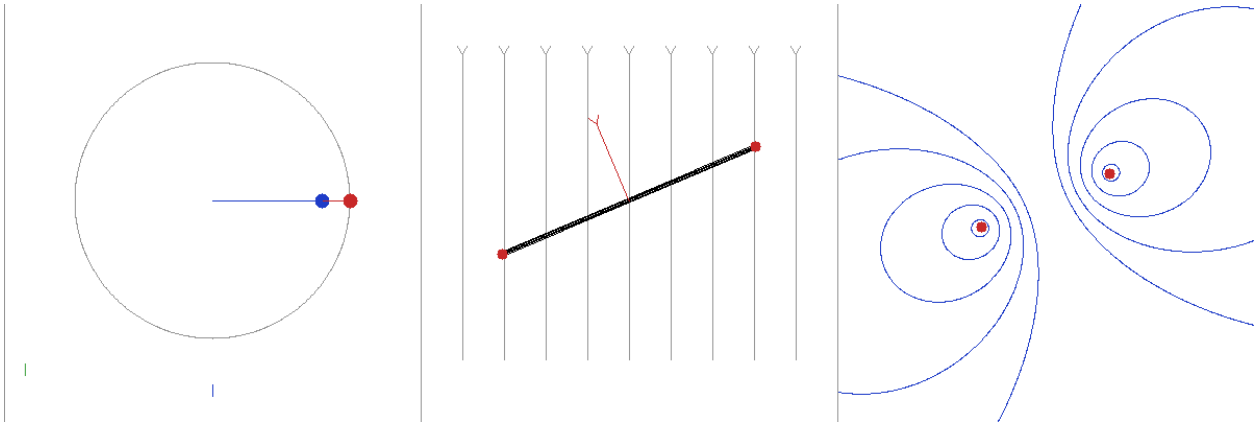


Figure 2. First frame of viz/ring_motion.gif (90 frames). Left: Kelvin's switch-on seen from above, tube marker red and fluid marker blue (angles scaled; the real ones are microradians), coil current as the green bar. Middle: the tilting ring edge-on (black bar, wire ends red) in the vertical field (grey arrows), its magnetic moment as the red arrow; the 0.05 rad swing drawn ×8. Right: its dipole field lines, which rock with it at the pendulum period. The animation is drawn from the solver's own time points.

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Validation

test	check	held to
TestKelvinTwin	moments −/+eΦ/2π, fluid at Larmor's rate, bench ledger closed, with and without the ring's self-inductance; against the Lorentz-force bench	1e-6; bench 1e-5, kinetic energy 2e-5
TestTiltPeriod	period against $4K(k)/\omega_0$; ring current conserved	2e-4; drift 1e-9
TestTiltHarmonics	3rd and 5th harmonics against the Jacobi series at $\theta_0 = 1$ rad; no even harmonics; small-angle 3rd = $\theta_0{}^2/64$	2e-4; 1e-3
TestElectronRing	$g\cdot(1 + Le^2/4\pi^2ma^2) = 1$; precession = g × Larmor; Compton ring below c; fitted ring carries the measured moment and is superluminal; every scenario above the size bound	1e-12; 1e-9
TestMasslessSpeed	$v/c = \pi/(2\alpha(\ln(8a/r) - 7/4))$ at a = 1 fm, $\hbar/m_e c$ and 0.1 nm	1e-9
TestFrequencies	$c/2\pi a = m_e c^2/h$ for the Compton ring; precession = $g\cdot eB/4\pi m$; fitted ring at the measured ESR; Kelvin ring libration period 19.869 s	1e-9; 1e-12; 1e-4; 0.01 s
TestFieldLinesClose	field lines of the ring current close	1e-9

Limits of the model

- The N element has one angle, so the ring cannot precess in 3-D; the Zeeman polarisation test stays with the 3-D bench of examples/kelvin_ring.
- No restoring force holds the ring's radius (the electrostatic hoop stress is unbalanced); a breathing mode would need a declared stiffness.
- Thin-ring formulas for L and C, with a core radius of a/100.

Status

H21: bounded. Step 1 (the twin test) passes; step 2 is computed: no ring with the electron's charge-to-mass ratio reaches g = 2 ($g \le 1$, Larmor precession eB/2m), and the massless ring that carries the measured moment at $S = \hbar/2$ needs v = 43.6 c at a = 8.9 fm. The standard prediction (Dirac + QED) is given beside every row. Tewari's space-vortex electron (H13) on the same bench gives g = 15/8.

Reproducing the results

```
go run ./examples/kron_electron -png viz/ring_field.png
go run ./examples/kron_electron -gif viz/ring_motion.gif
go test ./examples/kron_electron
KRONSPICE_EXPORT=1 go test ./examples/kron_electron -run TestExport
python examples/kron_electron/viz/make_report.py examples/kron_electron/kron_electron_report.pdf
```


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